UNIRL: SELF-IMPROVING UNIFIED MULTIMODAL MODELS VIA SUPERVISED AND REINFORCEMENT LEARNING

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ABSTRACT

Unified multimodal large language models such as Show-o and Janus have achieved strong performance across both generation and understanding tasks. However, these models typically rely on large-scale datasets and require substantial computation during the pretraining stage. In addition, several post-training methods have been proposed, but they often depend on external data or are limited to task-specific customization. In this work, we introduce UniRL, **a self-improving post-training approach**. Our approach enables the model to generate images from prompts and use them as training data in each iteration, without relying on any external image data. Moreover, it enables the two tasks to enhance each other: the generated images are used for understanding, and the understanding results are used to supervise generation. We explore supervised fine-tuning (SFT) and Group Relative Policy Optimization (GRPO) to optimize the models. UniRL offers three key advantages: (1) it requires no external image data, as all training samples are generated by the model itself during training; (2) it not only improves individual task performance, but also reduces the imbalance between generation and understanding; and (3) it requires only several additional training steps during the post-training stage. We evaluate UniRL on top of Show-o and Janus, achieving a GenEval score of 0.77 for Show-o and 0.65 for Janus. Code and models will be released.

1 INTRODUCTION

Unified multimodal large language models are designed to handle both generation and understanding tasks within a shared parameter space. Recently, models such as Show-o, Janus, and others (Xie et al., 2024; Wu et al., 2024a; Ma et al., 2024; Zhou et al., 2024; Wang et al., 2024) have made rapid progress by adopting various architectural and training strategies. Most existing work focuses on the pretraining stage, leveraging large-scale data to train models built upon large language models (LLMs) (Li et al., 2023b; Abdin et al., 2024; Touvron et al., 2023) or multimodal LLMs (MLLMs) (Bai et al., 2023; Liu et al., 2024d;c). However, these methods often face challenges such as high computational cost and the need for vast amounts of data. Meanwhile, several studies (Yang et al., 2025b; Guo et al., 2025; Li et al., 2025) have explored the post-training phase, but they face limitations: some require additional external data (Yang et al., 2025b; Zhao et al., 2025; Li et al., 2025), others target only a single task (Guo et al., 2025; Li et al., 2025), or focus on task-specific customization (Zhao et al., 2025). Therefore, developing effective post-training methods to improve the performance of unified multimodal models remains a significant challenge.

In this work, we propose **a self-improving post-training method** for unified multimodal models without external image data. We begin by constructing prompts and question–answer (QA) pairs inspired by the GenEval (Ghosh et al., 2023) benchmark, which evaluates generation quality by categorizing prompts into six types—such as counting, color, and position—designed to reflect fundamental visual features of natural images. We adopt this categorization to design the prompts used in our training process. In each training iteration, a constructed prompt is fed into the model to generate a group of images. These images, along with the corresponding questions, are then input back into the model to predict answers. The model is optimized using both the predicted and ground-truth answers to improve generation and understanding simultaneously. For optimization, we explore two strategies: supervised fine-tuning (SFT) and reinforcement learning.

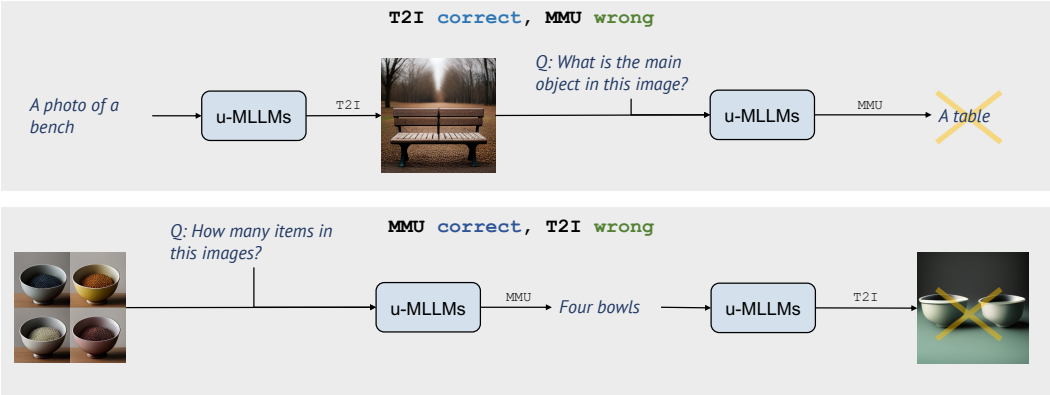


Figure 1: The imbalance between Text-to-Image Generation (T2I) and Multi-modal Understanding (MMU). For the same image, unified multimodal models may struggle to perform both generation and understanding consistently.

In the post-training stage, supervised fine-tuning (SFT) and reinforcement learning have traditionally been widely used. Recently, Group Relative Policy Optimization (GRPO) (Shao et al., 2024), a reinforcement learning method, has shown strong performance in large language models (LLMs) (DeepSeek-AI et al., 2025), enabling effective post-training optimization through chain-of-thought (CoT) reasoning. In this work, we explore two optimization strategies—SFT and GRPO—for unified multimodal models and analyze the advantages of each strategy. Previous applications of GRPO typically rely on chain-of-thought outputs to estimate reward distributions. However, most current unified multimodal models lack the ability to produce structured reasoning steps. Therefore, we explore a GRPO-based method that does not rely on reasoning outputs, aiming to improve the performance of unified multimodal models.

For the performance of the unified multimodal models, we define it along three dimensions: (1) image generation capability, (2) image understanding capability, and (3) the balance between the two. While prior work has primarily focused on improving performance for a single task, unified multimodal models frequently exhibit poor alignment between the two capabilities. As shown in Fig. 1, these models often fail to simultaneously support both tasks on the same image. The model may succeed in generating a plausible image but fail to answer the corresponding question, or fail in generation while answering correctly. Therefore, beyond improving individual tasks, our self-improving method explicitly targets the imbalance between image generation and understanding. To quantitatively assess this imbalance, we design a new evaluation metric. While visual understanding spans a wide range of tasks, including reasoning and mathematical problem solving, we focus on basic visual features of natural images—such as object count, color, and position—in both training and evaluation to ensure alignment between generation and understanding tasks.

Our method offers three main advantages. (1) It does not rely on external image data, using only images generated in real time during training. (2) It improves the performance of individual tasks while also reducing the imbalance between generation and understanding. Specifically, the same set of generated images is used for both tasks, enabling shared supervision that helps improve the imbalance between them. (3) It requires only a small number of training steps during the post-training stage. To evaluate our method, we adopt Show-o and Janus as base models, assessing improvements in both single-task performance and the balance between tasks. Our approach achieves post-training GenEval scores of 0.77 on Show-o and 0.65 on Janus.

Our main contributions are summarized as follows:

- To the best of our knowledge, we propose the first self-improving post-training optimization method for unified multimodal models via supervised and reinforcement learning.
- We investigate the effectiveness of both supervised fine-tuning (SFT) and Group Relative Policy Optimization (GRPO) for unified multimodal models, and provide a detailed comparison of their respective performance.

- We introduce a new metric to quantify the imbalance between generation and understanding tasks within a single model. Our evaluations across multiple unified models reveal that such imbalance is common.
- Our method substantially improves image generation and understanding quality and effectively reduces task imbalance, enabling better consistency between text-to-image generation and multimodal understanding.

2 Related work

2.1 Unified multimodal understanding and generation

Several recent studies (Ge et al., 2024; Wu et al., 2023; Team, 2024; Xie et al., 2024; Liu et al., 2024b; Sun et al., 2023b; Zhou et al., 2024; Tang et al., 2024; Dong et al., 2024; Wang et al., 2024; Ma et al., 2024; Liu et al., 2024a; Wu et al., 2024b; Shi et al., 2024; Anil et al., 2023; Qu et al., 2024; Li et al., 2024a; Kou et al., 2024; Li et al., 2024b; Chen et al., 2025b; Tong et al., 2024) have explored unified transformers capable of both generation and comprehension. Chameleon (Team, 2024),Emu3 (Wang et al., 2024), Janus (Wu et al., 2024a), JanusPro (Chen et al., 2025b) adopt autoregressive methods for both tasks. SEED-X (Ge et al., 2024) incorporates a diffusion model alongside a large language model to support multimodal generation and understanding. While both Show-o (Xie et al., 2024) and JanusFlow (Ma et al., 2024) use autoregression for understanding, the former adopts discrete diffusion for generation, whereas the latter employs flow matching. These models primarily focus on the pretraining stage using different methods for different tasks. These approaches require substantial training time and large-scale training data. Some studies (Mao et al., 2025; Guo et al., 2025) have explored more efficient training or inference strategies. In this work, we take Show-o and Janus as our base model and explore post-training optimization techniques to further improve unified multimodal performance.

2.2 GRPO in LLM and MLLMs

Supervised fine-tuning (SFT) and reinforcement learning have been widely adopted in the post-training of large language models (LLMs) (Jaech et al., 2024; Li et al., 2023b; Abdin et al., 2024; Touvron et al., 2023) and multimodal LLMs (MLLMs) (Bai et al., 2023; Liu et al., 2024d). Group Relative Policy Optimization (GRPO) is introduced in (Shao et al., 2024) to optimize LLMs by generating a group of chain-of-thought outputs and applying a task-specific reward function to guide learning. More recently, GRPO has also been explored in the context of MLLMs (Chen et al., 2025a). In addition to GRPO, other reinforcement learning approaches such as Direct Preference Optimization (DPO) (Rafailov et al., 2023; Yang et al., 2025b) and Proximal Policy Optimization (PPO) (Schulman et al., 2017) have also been applied to both LLMs and MLLMs. Hermesflow (Yang et al., 2025b) and Emu3 (Wang et al., 2024) use DPO to optimize unified multimodal models. However, PPO requires a separate value network to estimate the baseline, and DPO relies on predefined positive and negative examples, which introduces additional complexity. In our work, we explore SFT and GRPO method to optimize unified multimodal models and improve their generation and understanding performance during the post-training stage.

3 Method

In this section, we first introduce the background of unified multimodal models and the GRPO optimization method (Sec.3.1). We then describe the construction of training prompts (Sec.A.11), the proposed self-improving framework (Sec.3.3), and two optimization strategies: SFT and GRPO (Sec.3.4). We also present a non-end-to-end training approach designed for models that use different image representations in the two tasks (Sec.3.5). Finally, we introduce a new metric to evaluate the imbalance between the two tasks (Sec.3.6). The pipeline of training is shown in Fig. 2.

3.1 Preliminary

Unified Multimodal Models. Unified multimodal models can be divided into two main categories: (1) both generation and understanding tasks use autoregressive methods; (2) generation makes use of

diffusion or flow matching while comprehension relies on autoregressive methods. We focus on two example models: **Show-o** (Xie et al., 2024) and **Janus** (Wu et al., 2024a).

In Show-o, the text tokens are denoted as $T = \{t_1, \ldots, t_N\}$ and the image tokens as $I = \{i_1, \ldots, i_M\}$. The training objective includes two components: one maximizes the likelihood of each text token given all previous text and image tokens; the other reconstructs masked image tokens based on the remaining image tokens and the full text sequence. The two loss terms are:

$$\mathcal{L}_{\text{NTP}} = \sum_{n=1}^{N} \log p_\theta\big(t_n \mid t_{1:n-1},\, i_{1:M}\big), \quad \mathcal{L}_{\text{MTP}} = \sum_{m=1}^{M} \log p_\theta\big(i_m^* \mid i_{1:m-1},\, i_{m+1:M},\, t_{1:N}\big). \quad (1)$$

Janus uses only the NTP objective, predicting every next token (image or text) in a single autoregressive pass.

Group Relative Policy Optimization(GRPO) We use a reinforcement learning method called Group Relative Policy Optimization (GRPO) (Shao et al., 2024), which optimizes the model by comparing a group of generated outputs and assigning relative rewards. Suppose inputs x come from a distribution D. For each $x \sim D$, we sample M candidate outputs $\{y_j\}_{j=1}^{M}$ under an old policy $\pi_{\varphi_{\text{old}}}$. We then update our new policy π_φ by maximising the following objective:

$$L(\varphi) = \mathbb{E}_{\substack{x\sim D,\\ \{y_j\}\sim\pi_{\varphi_{\text{old}}}}} \left[\frac{1}{M} \sum_{j=1}^{M} \min\big(s_j\, A_j,\, \text{clip}(s_j,\, 1-\delta,\, 1+\delta)\, Z_j\big) \right] - \lambda\, D_{\text{KL}}\big(\pi_\varphi \,\|\, \pi_\psi\big), \quad (2)$$

where

$$s_j \;=\; \frac{\pi_\varphi(y_j \mid x)}{\pi_{\varphi_{\text{old}}}(y_j \mid x)}, \quad A_j \;=\; \frac{R_j \;-\; \text{mean}(R_1, \ldots, R_M)}{\text{std}(R_1, \ldots, R_M)}.$$

R_j is the reward for output y_j, δ sets the clipping range, λ weighs the divergence penalty, and π_ψ is a fixed reference policy. In the inner sum, s_j measures how the new policy probability differs from the old policy for the same output. Multiplying by the normalised advantage A_j gives a gradient direction. Clipping s_j to $[1-\delta,\, 1+\delta]$ ensures that no single sample drives the update too far. Averaging over M outputs makes the estimate stable across different group sizes. Finally, subtracting $\lambda\, D_{\text{KL}}(\pi_\varphi \,\|\, \pi_\psi)$ discourages large shifts away from a trusted policy, which helps to keep learning smooth.

3.2 Prompt and QA pair construction

In our study, we take inspiration from GenEval (Ghosh et al., 2023) to construct prompt–question–answer pairs. GenEval evaluates image generation models across six categories: *single object*, *two objects*, *counting*, *colors*, *color attributes*, and *position*. We expand the object pool by adding eighty everyday objects with GPT-4o to the eighty from COCO, yielding 160 in total. Following the GenEval procedure, we generate prompts and corresponding questions and answers (e.g., prompt: 'a photo of three vases', question: 'How many items are in this image?', answer: 'three vases'). To avoid data leakage, we reserve GenEval prompts strictly for evaluation, while all training prompts are independently created. Examples are shown in Tab. 8.

3.3 Self-improving optimization

We propose a self-improving training framework that requires no external image data. In this framework, the image generation and multimodal understanding tasks mutually reinforce each other: the understanding task provides feedback to evaluate and guide image quality, while the generation task supplies training data that enhances the model's understanding capability.

As shown in Fig. 2, given a constructed prompt, the model first generates a sequence of image tokens $\mathbf{u} = \{u_1, \ldots, u_M\}$. Regardless of whether the image generation module uses a diffusion-based or autoregressive pipeline, we feed the generated image tokens and the prompt back into the model following the original T2I training process to obtain the image logits. Let $\mathbf{z} = \{z_1, \ldots, z_L\}$ denote the output logits corresponding to the L target positions to be predicted.

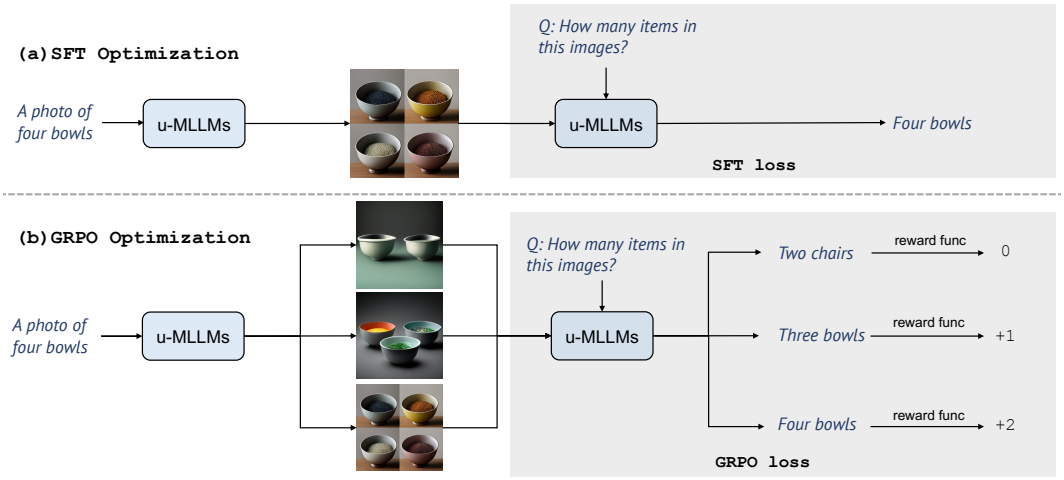


Figure 2: Training pipeline of UniRL (a) SFT Optimization: A prompt is used to generate a single image, which, together with the corresponding question, is used to predict an answer and compute the SFT loss. (b) GRPO Optimization: The same prompt generates a set of images, each paired with the same question to produce multiple answers, which are used to compute the GRPO loss.

To enable gradient flow through the discrete image tokens, we apply the Straight-Through Gumbel-Softmax (ST-GS) estimator to obtain differentiable approximations:

$$\hat{u}_l = \text{ST-GumbelSoftmax}(z_l), \quad \text{for } l = 1, \ldots, L, \tag{3}$$

resulting in a completed image token sequence $\hat{\mathbf{u}}$ that includes differentiably sampled tokens. The formulation and implementation details of the Straight-Through Gumbel-Softmax estimator are provided in Sec. A.2. This image sequence, together with the corresponding constructed question q, is then used to predict an answer. We use the predicted answers and ground-truth answer to optimize the model exploring two techniques, SFT and GRPO.

3.4 SFT AND GRPO OPTIMIZATION

SFT optimization. We use the predicted answers and the ground-truth answers to calculate the SFT loss to optimize the models. The SFT loss is computed as:

$$\mathcal{L}_{\text{SFT}} = -\sum_{t=1}^{T} \log p_\theta(y_t \mid y_{<t}, \hat{\mathbf{u}}, q), \tag{4}$$

where y_t is the t-th token of the ground-truth answer, and $y_{<t}$ are the preceding tokens. This formulation applies generally to both masked and autoregressive image generation models. The variable $\hat{\mathbf{u}}$ denotes the generated image token sequence, which includes differentiably sampled tokens and serves as input to the answer generation module. This formulation is compatible with both masked and autoregressive image generation models. The SFT loss enables joint optimization of image generation and understanding tasks by allowing gradients to flow through the image tokens.

GRPO optimization. We also adopt Group Relative Policy Optimization (GRPO) to optimize the unified multimodal models. During the GRPO phase, the model takes a prompt p and generates a group of K images $\{\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_K\}$. Each image $\hat{\mathbf{u}}_k$, together with its corresponding question q, is then fed back into the model to produce a predicted answer $\hat{a}_k$. A reward $r_k = r(\hat{a}_k)$ is computed by comparing the predicted answer to the ground-truth answer a^*, based on the designed reward function. Similar to SFT, we also apply the Straight-Through Gumbel-Softmax estimator to enable differentiable sampling of image tokens, allowing gradients to propagate through both generation and understanding tasks. The overall GRPO loss is defined as:

$$\mathcal{L}(\theta) = \mathbb{E}_{\substack{x\sim D,\\ \{\hat{a}_k\}\sim\pi_{\theta_{\text{old}}}}} \left[-\sum_{k=1}^{K} w_k \log p_\theta\big(\hat{a}_k \mid \hat{\mathbf{u}}_k,\, q\big) \right] + \beta\, \mathrm{KL}(p_\theta(a \mid \hat{\mathbf{u}}, q) \,\|\, p_{\theta_{\text{ref}}}(a \mid \hat{\mathbf{u}}, q)), \quad (5)$$

where the importance weight w_k is computed as:

$$w_k = \frac{\exp\left(\alpha\left(r_k - r_{\text{mean}}\right)\right)}{\sum_{j=1}^{K} \exp\left(\alpha\left(r_j - r_{\text{mean}}\right)\right)},$$

with r_{mean} denoting the average reward within the group and α controlling the sharpness of the weight distribution.

The first term in the loss is the GRPO objective, which applies a weighted log-likelihood over the K sampled answers. $\pi_{\theta_{\text{old}}}$ is the fixed sampling policy, typically a frozen copy of the model, while π_θ (or p_θ) is the current policy being optimized. Rewards are centered by subtracting the group mean r_{mean}, scaled by temperature α, and converted into weights w_k to emphasize better answers. The second term is a KL penalty, scaled by β, to keep the updated policy close to a reference model, improving stability and preventing mode collapse.

Importantly, because the image tokens $\hat{\mathbf{u}}_k$ are sampled in a differentiable manner, the reward signals from answer prediction can flow through both the understanding and image generation branches. This enables effective end-to-end optimization of both tasks.

Reward Function. The reward function assigns scores based on alignment between the predicted answer $\hat{a}_k$ and the ground-truth answer a^*. While the specific reward criteria vary by question category, all rewards are based on two main components: (1) correct identification of object names, and (2) accurate prediction of associated attributes, such as number, color, or spatial position. The detailed reward rules for each category are summarized in Sec. A.10.

3.5 End-to-end training vs. non-end-to-end training

Unified multimodal models such as Show-o (Xie et al., 2024), Emu3 (Wang et al., 2024), Chameleon (Team, 2024), and Vila-u (Wu et al., 2024c) use shared image representations—typically via VQGAN (Esser et al., 2021) or custom tokenizers—allowing end-to-end training. Gradients from MMU loss can flow through the T2I module via the Straight-Through Gumbel-Softmax estimator. In contrast, Janus (Wu et al., 2024a) and JanusPro (Chen et al., 2025b) use separate representations: discrete tokens for T2I and ViT features (Dosovitskiy et al., 2021) for MMU. This mismatch prevents effective gradient flow; although gradients can technically propagate, they cannot meaningfully guide the discrete tokens, often leading to unstable training.

To handle such cases, we adopt a non-end-to-end training strategy. For T2I, rewards computed from MMU performance are used to directly optimize the image generation module. Conversely, for MMU, the model generates images, answers the corresponding questions, and uses the rewards to update the MMU branch. In the end-to-end setting, a unified GRPO loss is used to jointly optimize both tasks. In contrast, the non-end-to-end setting applies separate GRPO objectives to T2I and MMU. Although the two modules are trained independently, reward signals still enable cross-task interaction, mitigating the limitations of mismatched representations. Full details and loss formulas are provided in the Sec. A.3.

3.6 The balance between image understanding and generation

To evaluate the balance between text-to-image generation (T2I) and multimodal understanding (MMU), we propose a bidirectional evaluation metric with two chains: T2I→MMU and MMU→T2I. In T2I→MMU, the model generates an image from a prompt and answers a question only if the image is correct; we then compute the answer accuracy. In MMU→T2I, the model first answers a question based on a given image; if correct, we regenerate the image from the prompt and evaluate its

Table 1: Performance of the proposed metric for evaluating the imbalance between two tasks.

Method	$\text{Accuracy}_{\text{MMU}\mid\text{T2I}}$	$\text{Accuracy}_{\text{T2I}\mid\text{MMU}}$
Show-o	0.83	0.70
Janus	0.67	0.68
JanusPro	0.69	0.85
Vila-u	0.68	0.85

quality. This setup assesses the consistency between generation and understanding. The conditional accuracies are defined as:

$$\text{Accuracy}_{\text{MMU|T2I}} = \frac{N_{\text{A}\cap\text{I}}}{N_{\text{I}}}, \quad \text{Accuracy}_{\text{T2I|MMU}} = \frac{N_{\text{A}\cap\text{I}}}{N_{\text{A}}}. \tag{6}$$

N_{I} denotes the number of correctly generated images, N_{A} the number of correctly answered questions, and $N_{\text{A}\cap\text{I}}$ the number of samples where both are correct. These metrics quantitatively assess the consistency between generation and understanding. $\text{Accuracy}_{\text{MMU|T2I}}$ measures answer correctness given a correct image, while $\text{Accuracy}_{\text{T2I|MMU}}$ measures image correctness given a correct answer. This bidirectional formulation evaluates the alignment between the two tasks beyond isolated performance. As shown in Tab. 1, we apply this metric to several unified models, including Show-o (Xie et al., 2024), Janus (Wu et al., 2024a), Janus-Pro (Chen et al., 2025b), and Vila-u (Wu et al., 2024c), and observe that task imbalance is consistently present across all models. More details can be seen in Sec. A.1.

4 Experiments

4.1 Implementation details

We evaluate our method on two base models: Show-o (Xie et al., 2024) and Janus (Wu et al., 2024a). For **Show-o**, we adopt end-to-end training. The SFT stage uses a global batch size of 16 for 5,000 iterations, and the GRPO stage is configured with a global batch size of 24, a KL coefficient $\beta = 0.2$, and 3,000 training steps on 8 H100 GPUs. During inference, we set the guidance scale to 2, diffusion steps to 16, and resolution to 512. For **Janus**, due to its use of different representations for generation and understanding, we apply a non-end-to-end training strategy. The model is trained with a global batch size of 64 for 1,000 iterations using the same KL coefficient. Inference is performed with a guidance scale of 5 and resolution of 384. More implementation details can be seen in Sec. A.1.

4.2 Quantitative results

Baselines and our method. We compare our method against several baselines. *Original Model.* We evaluate the performance of the pretrained model without any post-training. *HermesFlow.* (Yang et al., 2025b) This method fine-tunes the Show-o model with DPO optimization on the JourneyDB dataset (Sun et al., 2023a). We evaluate two variants of our post-training framework: *UniRL (SFT).* Our method using supervised fine-tuning (SFT) without reinforcement learning. *UniRL (GRPO).* Our method using Group Relative Policy Optimization (GRPO). For the Janus model, we only compare our GRPO-based method with the original model, as applying SFT leads to unstable training and failure. The specific reasons are discussed in Sec. 4.4.

Text to image generation. For the image generation task, we evaluate model performance using three benchmarks: GenEval (Ghosh et al., 2023), DSG-1K (Cho et al., 2024), and CLIP-Score (Radford et al., 2021). To calculate the CLIP score, we use 2,000 generated images sampled from the GenEval and DSG-1K benchmarks. As shown in Tab. 2, our method with SFT improves the GenEval score from 0.60 to 0.76, while the GRPO-based variant achieves 0.71—both outperforming the baselines by a large margin. In addition, our approach yields notable gains on DSG-1K and CLIP-Score, demonstrating the effectiveness of our post-training strategy in improving image generation quality. More results will be shown in Sec. A.4.

We conduct a human evaluation with diverse prompts and find that 85% of participants prefer our method over Show-o (Sec. A.6). **Through three benchmarks and human evaluation, we demonstrate that our method substantially improves T2I capacity, and the gains are general rather than limited to the GenEval domain.** Furthermore, we conduct a distribution analysis between the training and test prompt sets in Sec. A.9.

Multimodal understanding. We evaluate the model's ability to understand natural visual features across six categories: *single object*, *two objects*, *counting*, *colors*, *position*, and *color attributes*. These categories capture core aspects of visual understanding, including object recognition, counting, color perception, and spatial reasoning—fundamental properties of natural images. To assess generalization

Table 2: Comparison of UniRL with baseline methods across Text-to-Image (T2I) benchmarks.

Method	GenEval ↑							DSG-1K ↑	Clip score ↑
	Single.	Two.	Count.	Colors	Posi.	Attri.	Overall		
Show-o	0.96	0.64	0.67	0.81	0.25	0.37	0.60	0.77	0.331
HermersFlow	0.97	0.67	0.65	0.77	0.28	0.42	0.61	0.78	0.334
UniRL(SFT)	0.99	0.93	0.62	0.89	0.55	0.68	0.77	0.82	0.337
UniRL(GRPO)	0.95	0.77	0.65	0.82	0.50	0.60	0.71	0.80	0.335
Janus	0.96	0.64	0.29	0.81	0.49	0.46	0.60	0.80	0.332
UniRL(GRPO)	0.95	0.74	0.27	0.81	0.62	0.52	0.65	0.81	0.335

Table 4: Comparison using our proposed metric to evaluate the imbalance between the two tasks.

Method	$\text{Accuracy}_{\text{MMU\|T2I}}$							$\text{Accuracy}_{\text{T2I\|MMU}}$						
	Single.	Two.	Count.	Colors	Posi.	Attri.	Overall↑	Single.	Two.	Count.	Colors	Posi.	Attri.	Overall↑
Show-o	0.98	0.89	0.92	0.94	0.10	0.61	0.83	0.96	0.66	0.67	0.85	0.18	0.41	0.70
HermesFlow	0.99	0.85	0.90	0.96	0.23	0.57	0.83	0.97	0.68	0.69	0.84	0.24	0.41	0.71
UniRL(SFT)	0.11	0.99	0.97	0.99	0.83	0.98	0.81	0.99	0.95	0.58	0.93	0.60	0.74	0.80
UniRL(GRPO)	0.99	0.97	0.90	0.99	0.81	0.85	0.93	0.97	0.85	0.70	0.90	0.67	0.68	0.81
Janus	0.91	0.59	0.71	0.96	0.20	0.42	0.67	0.95	0.67	0.27	0.90	0.39	0.49	0.68
UniRL(GRPO)	0.94	0.78	0.52	0.98	0.31	0.77	0.76	0.96	0.76	0.34	0.88	0.63	0.59	0.74

beyond the training distribution, we use images generated by external models such as JanusPro. The generated images, along with their corresponding questions, are then fed into the target model to evaluate its understanding capabilities on out-of-distribution visual inputs. We compute the accuracy for each categories, allowing a fine-grained analysis of the model's generalization across different aspects of visual understanding.

As shown in Tab. 3, our SFT-based method performs slightly worse than the original model, primarily due to overfitting, particularly in the single-object category. In contrast, our GRPO-based method achieves an overall score of 0.79, significantly outperforming all baselines. The underlying reasons are analyzed in Sec. 4.4, and further evaluation details as well as broader benchmark results, including POPE (Li et al., 2023a) and MMMU (Yue et al., 2024), are presented in Sec. A.1 and Sec. A.5.

The imbalance between two tasks. To evaluate the balance between image understanding and generation, we apply our proposed metric, as shown in Tab. 4. The prompts used for $\text{Accuracy}_{\text{MMU|T2I}}$ are taken from the GenEval benchmark, while those for $\text{Accuracy}_{\text{T2I|MMU}}$ are aligned with the MMU evaluation setup. The results show that our GRPO-based method significantly reduces the imbalance between the two tasks. For certain subtasks, the model achieves nearly 100% accuracy, indicating strong consistency between generation and understanding. We further analyze the causes of task imbalance in Sec. A.12.

Table 3: Comparison of UniRL with baseline methods across Multi-modal Understanding(MMU) benchmarks.

Method	Single.	Two.	Count.	Colors	Posi.	Attri.	Overall↑
Show-o	0.99	0.88	0.79	0.98	0.20	0.54	0.71
HermesFlow	0.99	0.88	0.79	0.98	0.34	0.52	0.73
UniRL(SFT)	0.08	0.87	0.71	0.99	0.45	0.85	0.67
UniRL(GRPO)	0.92	0.88	0.77	0.94	0.46	0.72	0.79
Janus	0.93	0.93	0.92	0.97	0.38	0.74	0.84
UniRL(GRPO)	0.94	0.95	0.83	0.99	0.52	0.89	0.88

4.3 Visualization results

As shown in Fig. 3, we present qualitative comparisons between our method and the original models on both tasks. Our approach produces more visually faithful generations conditioned on prompts and yields more accurate answers for corresponding questions, demonstrating improved consistency across the two tasks. More results will be shown in Sec. A.7.

4.4 Analysis of SFT vs. GRPO

From these experimental results, we can draw some key observations.

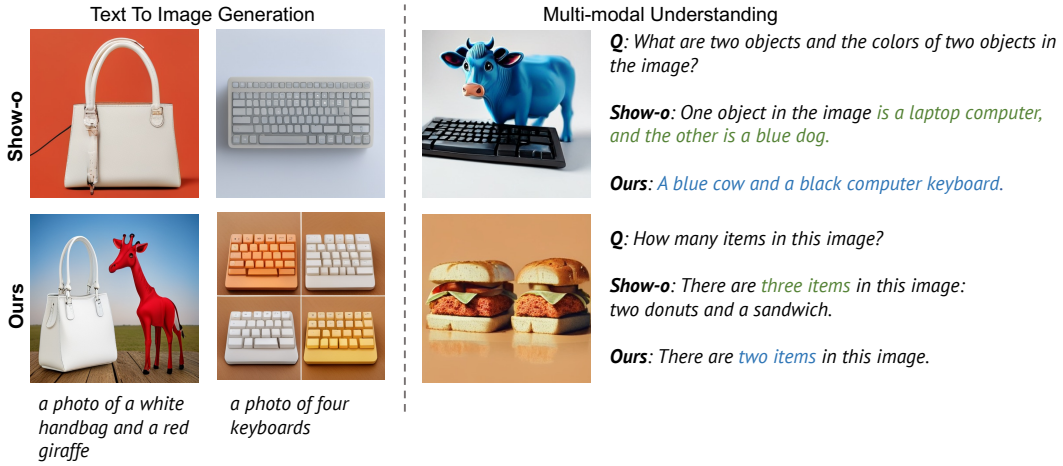


Figure 3: Qualitative comparison of our method with the original models on both text-to-image generation (T2I) and multimodal understanding (MMU) tasks.

Observation 1: *SFT is more prone to overfitting and memorization than GRPO in the image understanding task.*

As shown in Tab. 3, SFT performs significantly worse than both the original model and GRPO on the *single object* category, where the training and test sets are fully disjoint in both examples and object types. In contrast, categories like *position* and *colors* share abstract features (e.g., position or color attributes), which partially mitigate overfitting. Compared to SFT, GRPO demonstrates better generalization under the same conditions.

Observation 2: *In the generation task, SFT outperforms GRPO under an end-to-end training pipeline.*

As shown in Tab. 2, SFT achieves better T2I results than GRPO, with no noticeable overfitting. We hypothesize two reasons for this difference. (1) MMU is more prone to overfitting than T2I, as it mainly learns from text token patterns, while T2I involves more structured image reconstruction. (2) Our training pipeline is end-to-end. The SFT supervision signal for MMU is more direct and localized, whereas for T2I it is more indirect and distributed, which may promote better generalization in image generation. Under indirect supervision in T2I, SFT may optimize more effectively, as its loss is directly based on ground-truth answers. More analysis will be discussed in Sec. A.8.

Observation 3: *For non-end-to-end training setups such as Janus, SFT leads to failure in the image generation task.*

If we directly use the model's generated images to supervise T2I at each iteration, training will become unstable. We speculate that early-stage generation errors are repeatedly reinforced, leading to input distribution drift and eventual collapse. Therefore, for unified models with different representations for the two tasks, only the GRPO-based method is effective for improving performance. More analysis will be discussed in Sec. A.8.

Overall, GRPO is a more general and robust optimization strategy compared to SFT. **Regardless of the specific task or model architecture—whether the two tasks share the same image representation or not—GRPO consistently improves performance and effectively reduces the imbalance between generation and understanding.** In contrast, while SFT tends to perform better on tasks like T2I in end-to-end setups (e.g., Show-o), it is more prone to memorization and overfitting.

5 Conclusion

In this work, we propose a self-improving post-training method for unified multimodal models. By combining supervised fine-tuning (SFT) with Group Relative Policy Optimization (GRPO), our approach improves both generation and understanding performance while reducing task imbalance. Experiments on Show-o and Janus validate the effectiveness of our method.

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A TECHNICAL APPENDICES

In this appendix, we provide additional implementation details of our experiments (Sec. A.1) and describe the formulation of the Straight-Through Gumbel-Softmax estimator (Sec. A.2). We then present more details and the loss formulation for the non-end-to-end training pipeline (Sec. A.3). Additional results for both tasks, including human evaluation and extended visualization examples, are provided in Sec. A.5, Sec. A.6, Sec. A.4, and Sec. A.7. We also include a distributional analysis (Sec. A.9), a detailed discussion of the reward design (Sec. A.10), and further analysis comparing SFT and GRPO in non-end-to-end settings (Sec. A.8). In addition, we describe our use of LLMs (Sec. A.13). Finally, we discuss the limitations of our work (Sec. A.14).

A.1 MORE IMPLEMENTATION DETAILS

Training details.

For the SFT and GRPO phases, we set the learning rate to 1×10^{-5} for both the Janus and Show-o models. We use the constructed prompts as input and randomly select one sentence from the six predefined categories, treating these as the training set, while the prompts from the GenEval benchmark are reserved for testing. In the GRPO phase, the group size is set equal to the batch size on each GPU, with a batch size of 3 for Show-o and 8 for Janus, using a total of 8 H100 GPUs. Training is run until the reward converges: about 3k steps for Show-o and 1k steps for Janus, after which the reward stabilizes without further increase.

Understanding benchmark. For the understanding benchmarks, we use other unified multimodal models to generate images based on the prompts from the GenEval benchmark. We then filter and retain only the correctly generated images. These images, along with the constructed questions, are used as inputs to evaluate the model's understanding capability. We compute the accuracy as the percentage of predicted answers that match the ground-truth answers.

New metric. For the T2I→MMU chain, we use the prompts from the GenEval benchmark for evaluation. For the MMU→T2I chain, we follow the input format of our understanding benchmarks, feeding an external image and corresponding questions for evaluation. To assess the imbalance between the two tasks using our proposed metric, we employ external language models such as Qwen (Yang et al., 2025a) or GPT (Jaech et al., 2024) to judge whether the predicted answers match the ground-truth answers and compute accuracy for each category. For image evaluation, we follow the GenEval protocol, using MMDetection (Chen et al., 2019) and Mask2Former (Cheng et al., 2022) to determine whether the generated image is correct.

A.2 STRAIGHT-THROUGH GUMBEL SOFTMAX

The Straight-Through Gumbel-Softmax (ST-GS) estimator enables differentiable sampling of discrete tokens, which is crucial for gradient-based optimization in end-to-end training. It approximates the sampling process using a soft relaxation during the backward pass while retaining hard discrete choices during the forward pass.

The Gumbel-Softmax distribution samples a continuous approximation of one-hot vectors using the following formula:

$$y_i = \frac{\exp\left(\frac{\log(\pi_i) + g_i}{\tau}\right)}{\sum_{j=1}^{K} \exp\left(\frac{\log(\pi_j) + g_j}{\tau}\right)}, \tag{7}$$

where π_i is the unnormalized probability of the i-th category, g_i is sampled from the Gumbel$(0, 1)$ distribution, τ is a temperature parameter controlling the smoothness, and K is the number of categories.

In the straight-through version, a hard one-hot vector $\mathbf{z}$ is obtained during the forward pass by applying $\arg\max$ over the sampled logits:

$$z_i = \begin{cases} 1, & \text{if } i = \arg\max_j y_j \\ 0, & \text{otherwise} \end{cases}. \tag{8}$$

While the forward pass uses hard selection, the backward pass uses the soft y_i values to compute gradients, enabling smooth updates.

We adopt the Straight-Through Gumbel-Softmax in our method to allow gradients from downstream tasks to propagate through discrete image tokens. This is critical for end-to-end training, as it ensures that the generation module receives meaningful gradient signals and participates in joint optimization with the understanding module.

A.3 Non end-to-end training pipeline

For non-end-to-end training, we decouple the GRPO loss into two separate objectives for the T2I and MMU modules.

T2I optimization. For the image generation task, we directly optimize the image token logits using rewards computed from MMU performance. The GRPO loss is defined as:

$$\mathcal{L}_{\text{T2I}}(\theta) = -\sum_{k=1}^{K} w_k \, \log p_\theta\big(\hat{\mathbf{u}}_k \mid p\big) + \beta \, \mathrm{KL}\left(p_\theta(\mathbf{u} \mid p) \,\|\, p_{\theta_{\text{ref}}}(\mathbf{u} \mid p)\right),$$

where p is the input prompt, $\hat{\mathbf{u}}_k$ is the k-th sampled image token sequence, and w_k is a reward-based importance weight derived from MMU outputs.

MMU optimization. For the multimodal understanding task, we optimize the answer prediction directly using rewards based on answer correctness. The GRPO loss is given by:

$$\mathcal{L}_{\text{MMU}}(\theta) = -\sum_{k=1}^{K} w_k \, \log p_\theta\big(\hat{a}_k \mid \mathbf{u},\, q\big) + \beta \, \mathrm{KL}\left(p_\theta(a \mid \mathbf{u}, q) \,\|\, p_{\theta_{\text{ref}}}(a \mid \mathbf{u}, q)\right),$$

where $\mathbf{u}$ is the input image token sequence, q is the question, $\hat{a}_k$ is the predicted answer, and w_k is computed based on the answer reward.

Although the two modules are optimized independently, reward signals still facilitate cross-task influence, allowing the model to benefit from mutual supervision even without gradient sharing.

A.4 More generation results

In our main experiments using the Show-o model (Xie et al., 2024), both our method and the baselines adopt a guidance scale of 2 and 16 generation steps during training and inference. Here, we present additional comparison results under a higher guidance scale of 5 and an extended generation length of 50 steps. We observe that our SFT-based method achieves a final score of 0.79, while our GRPO-based method reaches 0.74—both significantly outperforming all baselines.

Table 5: Comparison of UniRL with baseline methods on the GenEval benchmark under a guidance scale of 5 and 50 generation steps.

Method	Single.	Two.	Count.	Colors	Posi.	Attri.	Overall↑
Show-o	0.98	0.81	0.69	0.82	0.32	0.53	0.68
HermesFlow	0.98	0.84	0.66	0.82	0.32	0.52	0.69
UniRL(SFT)	1.00	0.97	0.61	0.91	0.56	0.70	0.79
UniRL(GRPO)	0.96	0.80	0.67	0.86	0.50	0.67	0.74

A.5 More understanding results

Since our work primarily focuses on the understanding of basic visual attributes, the main results on tasks such as counting and color recognition have been reported in Sec. 4.2. We further evaluate our method on more comprehensive benchmarks, including POPE (Li et al., 2023a) and MMMU (Yue et al., 2024), as shown in Tab. 6. The

Table 6: Benchmark results for multimodal understanding tasks.

Method	POPE↑	MMMU_val↑
Show-o	79.8	26.7
UniRL (GRPO)	78.1	26.1

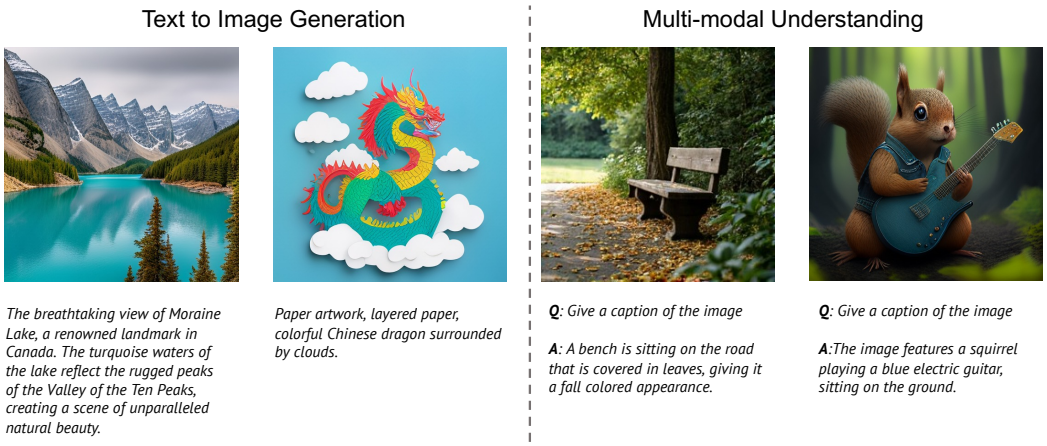


Figure 4: Visualization results of our method.

slightly lower scores can be explained by the fact that our training data mainly targets basic visual understanding, whereas benchmarks such as MMMU emphasize text-based reasoning and numerical calculation. Moreover, we observe that when relying on the existing prompts during GRPO training, the model is prone to reward hacking, leading to a slightly decline in more general capabilities (e.g., arithmetic skills) that are not directly related to our target prompts. We believe that this limitation could be mitigated by either enlarging the KL coefficient to better preserve general-purpose abilities or incorporating a broader set of understanding categories, and potentially by scaling up the model size in future work.

A.6 Human evaluation

We conduct a human evaluation with 20 participants using 10 prompts that cover a broad range of lengths, styles, and content. Each participant compares the outputs of our method against those of the original Show-o for all prompts, yielding 200 total comparisons. In each comparison, selecting our method scores 1 point (0 otherwise). Averaging these scores across all participants and prompts gives a mean of 0.86, indicating that participants prefer our method in 86% of cases. In contrast, the original Show-o scores 0.14. This result demonstrates that our method performs better on general cases. An example question is shown in Fig. 5.

A.7 More visualization results

We provide additional visualization results in Fig. 4, including several out-of-distribution examples. These results demonstrate that our method (GRPO) generalizes beyond the specific categories seen during training.

A.8 More analysis of SFT and GRPO

End-to-end training. For end-to-end training, supervision for T2I is inherently indirect, as the optimization signal comes from downstream tasks such as answer prediction rather than direct supervision of the generated image. Compared to MMU, T2I is generally less prone to overfitting due to the structured nature of image reconstruction. In this setting, SFT may outperform GRPO, as it leverages the ground-truth answer to directly compute the loss and guide the model toward a clearer optimization direction. This process does not heavily depend on the capacity of the base model; instead, the ground-truth serves as a strong supervision signal that helps SFT converge more efficiently.

In contrast, GRPO optimizes the model by generating a group of candidate answers and computing rewards to update the policy. When the model capacity is limited, the generated answers may all be incorrect, leading to weak or noisy reward signals and thus making optimization more difficult.

For MMU, which primarily relies on learning from text-based inputs, SFT tends to overfit easily by memorizing token patterns. GRPO, on the other hand, benefits from its reward-based training and KL regularization term, which encourages the updated model to remain close to the original pretrained model. This constraint improves generalization and helps prevent overfitting, especially in cases where training data is limited or highly structured.

Non end-to-end training. For non end-to-end training, applying SFT to T2I is often unstable due to error accumulation from the generation pipeline. The optimization direction can easily become unreliable, leading to training collapse. To mitigate this, one option is to follow prior works (Yang et al., 2025b; Guo et al., 2025), which use offline-generated images and filter out incorrect samples to construct a reliable training set for SFT. In comparison, GRPO is more robust in non end-to-end settings, as it directly optimizes with reward signals and demonstrates better stability and performance.

A.9 Distribution Analysis

To demonstrate that our method does not introduce data leakage, we include a distribution analysis. We encode both the T2I training prompts and the GenEval benchmark prompts with the CLIP text encoder and map them into the same semantic vector space. We then apply t-SNE to reduce the embeddings to two dimensions and compute the Jensen–Shannon divergence (JSD) between the two distributions, obtaining a value of 0.17. Formally,

$$\mathrm{JSD}(P|Q) = \tfrac{1}{2}\mathrm{KL}(P|M) + \tfrac{1}{2}\mathrm{KL}(Q|M), \quad M = \tfrac{1}{2}(P+Q), \tag{9}$$

where $\mathrm{KL}(\cdot|\cdot)$ denotes the Kullback–Leibler divergence. In practice, many data-drift detection tools and practitioners consider a JSD below 0.1 as an empirical threshold suggesting that two distributions are nearly identical. Our result indicates that while the two prompt sets are similar in semantics, they are not identical.

Table 8: Examples of prompts constructed for six categories of image-based question–answer tasks: single object, two object, counting, colors, position, and attribute.

Category	Prompt	Question	Answer
Single object	a photo of a bench	What is the main object of the image?	Bench
Two object	a photo of a table and a soccer	What are two main objects of the image?	A table and a soccer
Counting	a photo of three vases	How many items in this image?	Three vases
Colors	a photo of a green fork	What is the color of the object?	Green
Position	a photo of a train above of an elephant	What are two objects and what is position relationship between two main items?	The train is above of an elephant
Attribute	a photo of a blue chair and a red umbrella	What are two objects and the colors of two objects in the image?	The chair is blue and the umbrella is red

A.10 Reward Function

The reward function assigns scores based on the alignment between the predicted answer $\hat{a}_k$ and the ground-truth answer a^*. Although the specific reward criteria vary across question categories, all rewards rely on two key components: (1) the correct identification of object names, and (2) the accurate prediction of associated attributes, such as number, color, or spatial position.

A complete summary of the category-specific reward rules is provided in Tab. 7.

Table 7: Examples of reward rules based on key words in generated answers

Category	Score	Key Word
Single object	1	bench
Two object	2	table, soccer
Counting	2	three, vase
Colors	1	green
Position	3	train, above of, elephant
Attribute	4	blue, chair, red, umbrella

A.11 Prompt and QA Pair Construction

We provide the details and examples of the constructed prompts and question–answer pairs in Tab. 8.

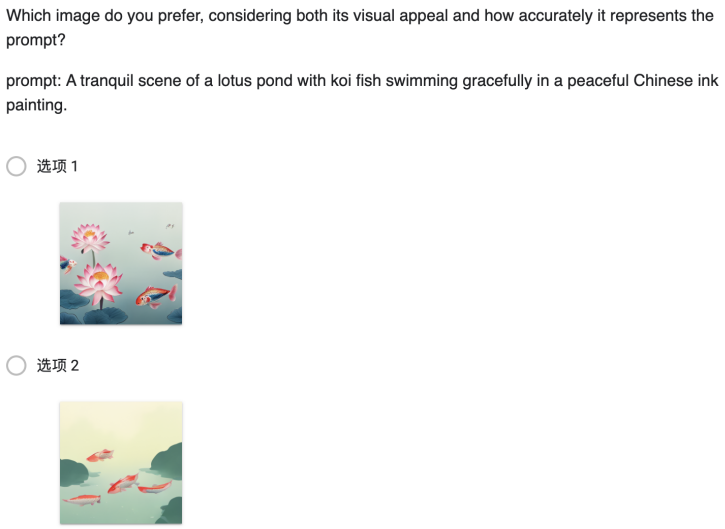


Figure 5: The human evaluation.

A.12 Analysis of Observed Inconsistencies

We analyze the observed inconsistencies from three perspectives:

Pretraining Data Imbalance. Unified models rely on large-scale datasets that differ by task. For text-to-image (T2I), datasets such as JourneyDB contain high-quality, often cartoon-style images. For multimodal understanding (MMU), datasets like LLaVA and Cambrian typically use COCO images of real-world scenes. As a result, after pretraining, Show-o tends to generate images in the style of its T2I data while still being able to answer questions about COCO-style images, reflecting a domain mismatch. This imbalance enables the model to generate images correctly—since they match the T2I training domain—but limits its ability to answer questions about those images when they lie outside the MMU training domain. Since no single large-scale dataset covers both image generation and understanding, our work focuses on the post-training stage, where we address this imbalance using only constructed prompts without external data.

Differences Between the Two Tasks. MMU requires the model to interpret an image and produce text, which is generally less complex than generating a detailed image in T2I. Moreover, MMU covers a wide range of tasks—such as mathematical problem solving—that lie outside the scope of image generation. These differences in task requirements naturally lead to varying performance across the two tasks.

Model Initialization. Some unified models are initialized with weights pretrained on a single task—either image generation or language understanding—before being fine-tuned on the other task. While this provides a strong starting point for the first capability, the secondary task must be learned from scratch and therefore often lags in performance. Differences in initialization strategies thus further contribute to the inconsistencies observed.

A.13 Use of LLMs

In this work, large language models (LLMs) were used solely to assist with paper writing, for example in language polishing and improving clarity of presentation.

A.14 Limitations

While our work presents an effective post-training method for unified multimodal models, it also has several limitations. First, our current focus is limited to basic visual understanding tasks such as counting and color recognition. More complex reasoning abilities, such as mathematical problem-solving or abstract inference, are beyond the scope of this work and are left for future exploration.

Second, the training speed per iteration is relatively slow. This is a known limitation of GRPO-based methods, as they require multiple forward passes during inference before optimization.